I (a) How the weight check works

1. The model reports one learned weight per compartment.



2. Hide compartments in every combination: $2^8 = 256$ sets in BLCA, $2^9 = 512$ in BRCA, each scored exactly.



3. Score every set twice: by each patient's risk score (what the model uses, no outcome) and by the out-of-bag C-index (what carries outcome signal). Shapley values share the credit.

risk score

C-index vs outcome

4. Compare the weights with what the model uses, and what the model uses with outcome.

weights ↔ use
faithful?

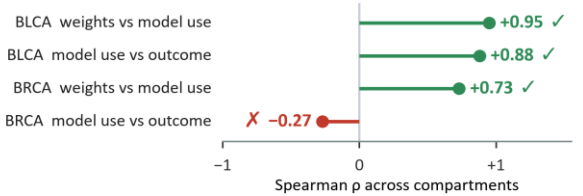
use ↔ outcome
prognostic?

STRESS TESTS

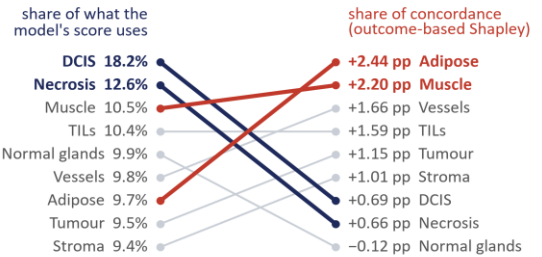
Repeat on hospitals the model never saw and with removed compartments masked out; plant a known signal in one compartment and ask which measure ranks it first.

(b) What the audit found

Faithful in both cohorts, prognostic only in BLCA



BRCA: what the model uses is not what predicts outcome



A known signal planted in one compartment

